1. Literature Review

1.1 Job Recommendation Systems

The importance of recommendation systems for online recruitment platforms lies in matching qualified candidates with relevant job opportunities [1], [2]. Traditional recommendation systems mainly focused on comparing resumes and job descriptions based on lexical similarities [1].

Recommendation systems commonly utilize text representation techniques such as TF-IDF, Word2Vec, and Doc2Vec [2]. Although these approaches can identify textual similarities between documents, they are not effective in capturing contextual relationships between resumes and job descriptions.

Advances in Natural Language Processing (NLP) and deep learning technologies have significantly improved the performance of modern recommendation systems [1], [2]. Transformer-based architectures such as BERT and SBERT enable systems to identify semantic similarities and contextual dependencies between resumes and job descriptions more accurately [1], [5].

One example of a modern recommendation system is CareerBERT, where resumes and occupations are represented within a shared embedding space to measure semantic similarity between candidate profiles and job descriptions [2]. Research results indicate that Transformer-based recommendation models outperform several traditional embedding approaches in job recommendation tasks [2].

Another example is JobFormer, which employs a semantic-enhanced Transformer architecture to capture contextual dependencies between resumes and job descriptions [1]. The framework consists of an encoder-decoder architecture, a two-stage recommendation pipeline, and a skill-aware recommendation mechanism [1].

Overall, modern recommendation systems increasingly rely on semantic embeddings, contextual analysis, and personalized recommendation strategies to improve recommendation quality and better align candidate profiles with labor market requirements [1], [2].

1.2 Application of Natural Language Processing in Recruitment

Natural Language Processing (NLP) is critical for modern recruiting tools due to its ability to automate the parsing of resumes and job descriptions. Standard keyword-based recruiting tools usually focus on lexical matching and cannot identify the context-based relations between candidate's experience and job requirements.

The use of NLP allows recruitment systems to parse the unstructured textual information and transform it into structured data that can be further processed. Modern recruiting tools apply natural language processing methods for parsing resumes, extracting skills, classifying occupations, and semantic matching [3].

Modern NLP tools for analyzing recruitment-related texts mainly leverage semantic embeddings, Transformer architectures, and large language models to process contextual dependencies. Such approaches increase semantic precision in the process of skill extraction and help identify related skills even in cases when the wording varies between resumes and job descriptions. One of the examples is ESCOX that leverages NLP to extract occupations and skills from unstructured texts using semantic similarity and alignment to ESCO taxonomy [3]. The system leverages semantic and context-based embedding techniques based on Transformer architecture and applies cosine similarity [3].

As stated in the ESCOX study, semantic extraction systems for recruitment purposes may prove useful for various tasks like labor market analysis, skill gap detection, workforce planning, and recommendation systems [3]. Also, NLP methods are instrumental in the automated processing of large amounts of recruiting data.

In conclusion, with NLP technologies, recruitment tools gain the ability to perform automated semantic parsing of unstructured recruitment data.

1.3 Semantic Matching and Embedding Models

Conventional recruiting systems tend to use keyword matching methods that compare exact matches of terms used in resumes and job descriptions [4]. Such systems usually fail due to discrepancies in terminology used by both parties to describe their respective qualifications and skills [4].

In order to overcome these drawbacks, modern recruitment systems employ semantic matching approaches based on embedding models. Contrary to the keyword-based approaches which see documents as sequences of isolated terms, embeddings capture the context and semantic connections between sentences and words represented by numerical vectors [5]. One of the widely used architectural solutions for embedding-based semantic document retrieval is Sentence-BERT [5]. Sentence-BERT generates sentence embeddings that enable quick search for semantic similarities between documents without the need for cross-encoders [5].

As far as semantic matching goes, cosine similarity measures are used to calculate angular distances between vector representations [4]. Semantically similar documents are closer in the embedding space and, thus, help recommend systems match candidates and job positions, even if their respective terminologies are different [4]. Many contemporary recruiting systems apply

vector search algorithms, where resumes and job descriptions are stored in vector indexes [4]. Candidate matching is performed based on finding semantically similar vector representations rather than comparing keywords [4].

Recent research proves that dense semantic matching algorithms can provide better results than keyword-based algorithms, such as BM25, for job matching tasks [4]. Moreover, Transformer embedding models trained with contrastive learning and hard negative mining produce better candidate matching and vacancy recommendations [4].

Another direction in modern research involves designing domain-specific architectures for recruiting systems. For example, Resume2Vec is an architecture designed to create a joint embedding space for mapping resumes and job descriptions into semantically comparable vector representations [6].

1.4 Large Language Models for Recruitment Systems

Large Language Models (LLMs) are extensively used in recruitment systems for such functions as resume filtering, candidate ranking, resume personalization, and automated resume evaluation [7], [8]. Transformer-based architectures allow recruitment platforms to handle vast amounts of recruitment data more effectively than rule-based methods [7]. Furthermore, LLMs can identify semantic relations between candidates' skills and work experience on one side and jobs' needs on another allowing semantic analysis of professional credentials [7].

One application of LLMs to recruitment is resume personalization. Modern resume personalization systems based on LLMs allow job seekers to customize their resumes according to particular job offers changing wording, highlighting relevant skills, and aligning their resumes with the job description better [8]. These tools help job seekers tailor their resumes specifically for ATS [8].

Automated feedback generation is another example of the use of language models for recruitment. Language models can look through the resumes and provide recommendations regarding formatting, skill gaps, use of keywords, and elaboration on professional descriptions [8]. Prompt engineering is yet another crucial aspect of using LLMs in recruitment systems. Recent research suggests that chain-of-thought prompting, as well as task-specific prompting, can contribute to consistent resume evaluation by language models [7]. However, there are some critical drawbacks associated with the application of LLMs for recruiting purposes. LLM-generated resume assessments can considerably differ from those made by human recruiters because of hallucination problems, inconsistencies in reasoning, and differences in scoring [7]. Moreover, there are also some demographic biases concerning gender and ethnicity in language models when conducting candidate evaluations [7].

For these reasons, recent scholarly studies recommend treating LLMs as decision support systems and not relying entirely on machine-generated assessment results [7]. Approaches involving human participation, or human-in-the-loop frameworks, are considered much more trustworthy because they involve both machine processing and human review [7].

1.5 Chapter Summary

This chapter analyzed the theoretical foundations and recent advances in the field of intelligent personnel selection systems. Recommendation systems, Natural Language Processing algorithms, semantic matching algorithms, embedding models, and utilization of large language models were considered.

The research demonstrates, that based on the reviewed literature, the most recent advances in the field of recruitment technologies use semantics and Transformer architecture rather than the keyword-based approach. Embedding models and semantic search allow for finding connections between candidate information and job positions' descriptions.

The literature review also showed that Large Language Models could be used in the fields of resume customization, candidate rating, and automated feedback generation. However, according to the reviewed literature, LLM-based recruitment systems have several major disadvantages including hallucinations, inconsistent logic, and demographics bias.

Overall, it is clear that integration of the semantic recommendation systems with Large Language Models could become an important direction for intelligent recruitment support system development. Results obtained in this chapter are vital for further consideration of system creation and development issues in following chapters.

References

1. Z. Guan, J.-Q. Yang, Y. Yang, H. Zhu, W. Li, and H. Xiong, *"JobFormer: Skill-Aware Job Recommendation with Semantic-Enhanced Transformer,"* ACM Transactions on Knowledge Discovery from Data, vol. 19, no. 1, p. 18, 2024. DOI: 10.1145/3701735

2. J. Rosenberger, L. Wolfrum, S. Weinzierl, M. Kraus, and P. Zschech, *"CareerBERT: Matching resumes to ESCO jobs in a shared embedding space for generic job recommendations,"* Expert Systems with Applications, vol. 275, p. 127043, 2025. DOI: 10.1016/j.eswa.2025.127043

3. D. C. Kavargyris, K. Georgiou, E. Papaioannou, K. Petrakis, N. Mittas, and L. Angelis, *"ESCOX: A tool for skill and occupation extraction using LLMs from unstructured text,"* Software Impacts, vol. 25, p. 100772, 2025. DOI: 10.1016/j.simpa.2025.100772

4. X. Yu, R. Xu, C. Xue, J. Zhang, X. Ma, and Z. Yu, *"ConFit v2: Improving Resume-Job Matching using Hypothetical Resume Embedding and Runner-Up Hard-Negative Mining,"* Findings of the Association for Computational Linguistics: ACL 2025. DOI: 10.18653/v1/2025.findings-acl.661

5. N. Reimers and I. Gurevych, *"Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,"* Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2019. DOI: 10.48550/arXiv.1908.10084

6. R. V. K. Bevara, P. Kumar, P. S. Reddy, et al., *"Resume2Vec: Transforming Applicant Tracking Systems with Intelligent Resume Embeddings for Precise Candidate Matching,"* Electronics, vol. 14, no. 4, p. 794, 2025. DOI: 10.3390/electronics14040794

7. S. Vaishampayan, H. Leary, Y. Alebachew, et al., *"Human and LLM-Based Resume Matching: An Observational Study,"* Findings of the Association for Computational Linguistics: NAACL 2025. DOI: 10.18653/v1/2025.findings-naacl.270

8. D. Awasthi, S. Majumder, J. Kim, et al., *"Supporting Job Seekers with LLM-Driven Resume Feedback,"* Proceedings of the 7th ACM Conference on Conversational User Interfaces (CUI 2025). DOI: 10.1145/3719160.3737622